Yucheng (Cecilia) Zhang

737-343-4836 | ceciliayczhang@gmail.com | Portfolio | LinkedIn | GitHub

Education

Northwestern University, M.S. Computer Science — Sep. 2026 – May 2028, Evanston, IL

The University of Texas at Austin, B.S. Mathematics; CS Minor; GPA: 3.74/4.00 — Aug. 2023 – May 2026, Austin, TX

Relevant Coursework: Stochastic Processes, Probability and Statistics, Real Analysis, Data Analytics, Database Systems, Artificial Intelligence

Technical Skills

Quant/ML: Machine Learning, statistical modeling, predictive modeling, feature engineering, model validation, backtesting, ROC/AUC, classification

Programming: Python, SQL, R, MATLAB, SAS; Pandas, NumPy, Scikit-learn, TensorFlow, Power BI, Git

Data/Research: SQL Server, Oracle, PostgreSQL, transaction data, SEC filings, RNA-seq, EEG/fNIRS, data cleaning, experiment analysis

Financial Analytics: Credit risk modeling, fraud analytics, transaction monitoring, risk segmentation, customer risk scoring

Experience

Enterprise Risk Analyst Intern — May 2025 – Dec. 2025

Toyota Industries Commercial Finance | Python, SQL Server, Azure, Power BI — Dallas, TX

- Developed an end-to-end credit risk analytics pipeline integrating PayNet inquiries and internal loan servicing data through SQL Server ETL, credit-feature engineering, Python modeling, and validation.
- Engineered borrower, contract, and asset-level features including loan-to-value ratio, account tenure, inquiry patterns, asset segments, and loan servicing attributes.
- Validated model performance with ROC/AUC, confusion matrices, and feature importance, achieving **AUC > 0.82** and improving high-risk case identification by approximately **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard segmentation updates.
- Automated monthly model refreshes with parameterized SQL and Python workflows connected to Power BI dashboards, reducing manual reporting effort by 10% and supporting recurring portfolio risk monitoring.

Financial Data Development Engineer – Fraud Detection — Jul. 2024 – Aug. 2024

Caitong Securities Ltd | SQL, Oracle, Stored Procedures — Hangzhou, China

- Designed SQL ETL pipelines processing **10M+ daily transactions** with window functions, subqueries, and stored procedures, improving extraction efficiency by 5%.
- Implemented automated fraud-detection logic with parameterized thresholds and modular stored procedures, reducing false positives by 2% in transaction-monitoring workflows.
- Developed a customer risk scoring engine by integrating transactional and behavioral data streams for fraud analytics, reviewer triage, and AML-related review workflows.

Projects

CodeAssist – AI Feedback System | GitHub — Aug. 2025 – Present

University of Texas at Austin | Flask, React, PostgreSQL, OpenAI — Austin, TX

- Built a full-stack AI feedback system for programming courses, connecting code editing, file uploads, submissions, and AI chat feedback for debugging, complexity reasoning, and test-case correctness.
- Developed Flask REST APIs and PostgreSQL models for AI feedback settings, student submissions, request limits, cooldown policies, JSON serialization, and role-based access.
- Analyzed Docker-based autograder bottlenecks and scoped async grading with job status tracking, timing metrics, timeout handling, and isolated submission/result paths.

Researcher & TA – Biological Data Analytics — Jan. 2024 – Dec. 2024

University of Texas at Austin | R, Linux, AWS, Docker — Austin, TX

- Applied statistical modeling (differential-expression analysis, PCA, permutation-based FDR) to 250GB of RNA-seq data as a research TA, identifying 48 key genes across reproducible R/Linux pipelines.
- Automated normalization, PCA, and volcano-plot workflows in R/ggplot2, improving interpretation efficiency by 15% for time-series biological datasets.
- Applied Salmon, RSEM, Trinity, and samtools for transcript quantification, achieving a **98.7% mapping rate** across reproducible datasets.

Neuroengineering Lab – Machine Learning Research Assistant — Jan. 2023 – May 2024

Python, MATLAB, MNE, RF, SVM, KNN, XGBoost — Remote

- Processed **35GB** of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, statistical testing, and emotion-recognition model evaluation.
- Applied statistical tests and feature-extraction methods to evaluate cognitive impairment patterns and emotion-recognition signals across experimental conditions.
- Evaluated RF, SVM, KNN, and XGBoost models across experimental conditions, contributing quantitative analysis used in peer-reviewed publications.

FinChat – AI-Powered Financial Analysis System — Jan. 2023 – May 2024

Southern Methodist University | Python, FinGPT, LangChain, CrewAI — Dallas, TX (Remote Hybrid)

- Built ETL pipelines to extract and process 100+ SEC filings from EDGAR, structuring unstructured financial data into a scalable data lake for metric extraction and trend detection.
- Implemented LangChain and CrewAI workflows for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time for filing analysis.
- Connected filing preprocessing, retrieval, metric extraction, and financial-question workflows into a reusable prototype for comparing companies across reporting periods.